

Priyanka Gugulothu

Data Analyst

MO • gugulothupriyanka98@gmail.com • +1 (314)-906-4252 • [Linkedin](#)

SUMMARY

Data Analyst with over 4 years of experience in data analytics, machine learning, **ETL**, and business intelligence across consulting, finance, and healthcare domains. Expert in **SQL, Python, R, Power BI, and Tableau** to develop data-driven solutions for predictive analytics, fraud detection, patient outcomes, and process automation. Proficient in cloud platforms (**Azure, AWS**), **Big Data (Spark, Hadoop)**, and **AI/ML (Scikit-learn, NLP, BERT)** to enhance data pipelines, automate workflows, and support clinical and operational decision-making. Proficient in data warehousing, compliance (**SOX, GRC, HIPAA**), and **Agile methodologies (Jira, UAT, Scrum)**, delivering scalable analytics solutions for global teams in healthcare and other sectors.

TECHNICAL SKILLS

- | | |
|--|--|
| • Programming Languages | Python, R, SQL |
| • Database Management | MySQL, PostgreSQL, MongoDB, Azure MySQL, Data Warehousing |
| • Data Processing & Analytics | Pandas, NumPy, PyTorch, SAS, Big Data, ETL (Clover ETL, SQL-based Extraction), Data Cleaning |
| • Data Visualization | Power BI, Tableau, ggplot2, Matplotlib, Seaborn, R Shiny, MS Excel |
| • Machine Learning & AI | Scikit-learn, Azure AI/ML, BERT, ROBERTA, NLP, Supervised & Unsupervised ML, Cosine Similarity |
| • ETL & Data Migration | Clover ETL, SQL-based Data Extraction, Azure Databricks, Data Pipelines |
| • Cloud & Technologies | AWS (S3, EC2, RDS), Azure AI/ML, SaaS, Docker, Kubernetes |
| • Big Data & Streaming | Apache Spark, Hadoop, Data Streaming, Real-time Analytics |
| • Automation & Workflow | Power Automate, Microsoft Lists, Jenkins, GitLab CI/CD |

PROFESSIONAL EXPERIENCE

BCBS

Data Analyst

MO

Mar 2024 – Present

- Analyzed **Python (Pandas, NumPy)** and **SQL** to analyze complex datasets related to claims, member utilization, and health outcomes. Collaborated with cross-functional teams to deliver actionable insights, enhancing operational efficiency for 10+ provider network and care management projects.
- Developed predictive machine learning models using **PyTorch and R** to forecast high-risk member populations and healthcare costs. Optimized algorithms to reduce forecast errors by 15% for 5 enterprise clients, improving care coordination and cost management.
- Built **ETL pipelines** with **SAS and Python (Pandas)** to automate aggregation of claims, enrollment, and provider data. Cut manual processing time by 20 hours monthly while maintaining **HIPAA compliance** and data integrity.
- Produced clear visualizations using **Tableau and Matplotlib** to present insights on healthcare metrics like cost trends and readmission rates to non-technical stakeholders. Accelerated decision-making for 8 care coordination and network management teams.
- Created interactive dashboards and automated reports using **Excel (Pivot Tables, VLOOKUP)** to streamline analysis of claims and provider performance. Saved 15+ hours weekly on manual data compilation for analytics teams.
- Coordinated with **Agile teams** to deploy scalable data solutions for population health and provider analytics. Demonstrated teamwork to shorten project timelines by an average of 2 weeks, meeting regulatory deadlines.
- Enhanced **SQL queries** and database structures to streamline retrieval of large-scale claims and member data. Reduced runtime by 1.5 hours per cycle for high-priority analytics requests, enabling timely reporting.
- Combined **R (ggplot2)** with **Python** workflows to standardize reporting on key metrics like **HEDIS** measures and cost per member. Improved analytical consistency for 12+ stakeholders, supporting audit readiness.

FINLOCKER

AI/ML Engineer

St. Louis, MO

Aug 2022 – Dec 2023

- Designed scalable machine learning models using **Python and TensorFlow**, collaborating with data teams to improve loan approval prediction accuracy, reducing processing time by 3 hours per application cycle.
- Developed data preprocessing pipelines with scikit-learn, enhanced data quality, which minimized preprocessing errors by 200 instances monthly.
- Collaborated with **DevOps** to deploy **ML models** on **AWS**, integrating **CI/CD workflows**, improving deployment efficiency, and reducing release timelines from 2 weeks to 3 days.
- Evaluated model performance using **PyTorch and cross-validation techniques**, applying detail to achieve a 98% accuracy rate in fraud detection, exceeding prior benchmarks.
- Optimized cloud infrastructure with **Docker and Kubernetes**, adapting to user loads to support 1 million daily API requests without latency spikes
- Led code reviews and **Agile sprint** planning, demonstrating leadership to accelerate bug resolution, cutting issue resolution time from 48 hours to 12 hours.

- Implemented **NLP solutions with spaCy** to automate customer support ticket categorization, boosting user satisfaction scores by 1.5 points on a 5-point scale.
- Conducted research on graph-based **ML algorithms**, and proposed a recommendation engine later integrated into the product roadmap.
- Automated **ETL pipelines** using **Apache Airflow and Spark**, demonstrating accelerated insights delivery from 8 hours to 1 hour per batch job.
- Mentored 3 junior engineers on **ML best practices**, fostering empathy to elevate team productivity and deliver 3 projects 10 days ahead of deadlines.

UNIVERSITY OF MISSOURI

St. Louis, MO

Graduate Teaching Assistant

Feb 2022 – Dec 2023

- Taught **Python, R, and SQL** concepts, mentoring 300+ students in statistical modeling and machine learning.
- Led business analytics projects using **Tableau** providing data-driven solutions for local businesses.
- Conducted hands-on workshops on Big Data tools like **Hadoop & Spark**, enhancing student project capabilities.
- Supervised capstone projects integrating **Azure AI/ML** for real-world predictive analytics solutions

CIPLA

India

Data Analyst

Aug 2020 – Dec 2021

- Analyzed pharmaceutical and clinical trial datasets using **SQL, Python (Pandas, NumPy), and Excel** to uncover drug efficacy patterns, reducing reporting time by 2 hours weekly.
- Designed interactive dashboards in **Power BI** to present clinical and sales KPIs, aiding 15+ departments in data-driven decision-making across therapy areas.
- Assessed and improved data integrity in clinical and manufacturing datasets using **ERwin and SQL**, ensuring 20% higher accuracy and **GDPR compliance**.
- Built predictive models in **Python (Scikit-learn)** to forecast drug demand and optimize supply chain variables, cutting annual operational costs by \$50K.
- Partnered with **R&D**, regulatory, and marketing teams via **Jira and Confluence** to capture analytical requirements, accelerating project delivery by 3 weeks.
- Automated ETL pipelines for patient adherence and inventory datasets using **Apache Airflow and Python**, reducing manual intervention by 8 hours weekly.
- Delivered strategic insights via data visualizations (**Matplotlib, Seaborn**) and **PowerPoint**, supporting executive decisions that increased therapeutic sales by \$1.2M.
- Conducted statistical root cause analysis with **R and SPSS** to identify compliance gaps and recurring data issues, resolving 15+ inconsistencies monthly.
- Guided junior analysts in preparing regulatory-grade documentation and using **Git** for version control, reducing onboarding time by 10 days.
- Managed pharmaceutical data storage using **AWS S3**, reducing costs by \$20K annually while enhancing scalability for global trials.

EDUCATION

UNIVERSITY OF MISSOURI – ST. LOUIS (UMSL)

Missouri, USA

Master of Science (M.S.) in Information Systems and Technology

Jan 2022 – Dec 2023

NATIONAL INSTITUTE OF TECHNOLOGY (NIT)

Telangana, India

Bachelor of Science (B.S.) in Biotechnology

Aug 2016 – Jul 2020

PROJECTS

Prediction of AI vs Human-Written Articles

- Collected and processed **Buzz feed and CNET articles** using **Selenium**, structuring data in **Excel & CSV** formats for analysis.
- Applied **VLOOKUP & Pivot Tables** to detect duplicate content and predict authorship with 80% accuracy, distinguishing AI-generated and human-written articles.

YouTube Analytics on AI News Anchors

- Integrated **YouTube API with R Programming**, extracting subscriber, like, and view data for comparative analysis.
- Conducted trend analysis using **ggplot2**, achieving 80% similarity in viewer engagement patterns between **AI and human news anchors**.

Twitter Data Analytics

- Collected Twitter data via **API**, applying **supervised machine learning (Linear Regression)** to predict retweet counts.

Data Warehouse Design for a Magazine Publisher

- Formulated **Entity-Relationship Diagrams (ERD) & Data Flow Diagrams (DFD)** using **Lucidchart**, mapping an optimized **data model** for magazine publishing.